

Soumyadeep Nath

Comprehensive Portfolio Synthesis, Technical Specification, & Agentic Workflow Profile

1. Personal & Professional Profile

My professional identity represents a highly structured, deliberate synthesis of advanced humanities methodologies and quantitative engineering systems. Holding a Master of Arts and a Bachelor of Arts with Honours in English Literature from Presidency University, Kolkata, I have engineered a systematic career pivot into data science and artificial intelligence through the Executive PG Programme in Data Science & AI at IIT Roorkee. Rather than minimizing this non-traditional background, I actively foreground the humanities-to-data-science transition as a major competitive advantage. This dual foundation enables the bridging of technical execution and stakeholder communication, translating complex statistical modeling and algorithmic architectures into clear narratives.

The Cognitive Humanities-to-Data Transition: The causal relationship between literary training and quantitative analysis manifests in my approach to unstructured data and system specifications. The study of literature demands the parsing of complex, highly unstructured semantic networks, identifying latent thematic patterns, and establishing structural rules. When translated into technical systems, this cognitive training yields a natural aptitude for advanced feature engineering, taxonomy-driven extraction, and strict semantic validation. I do not view code as a mere sequence of instructions, but as a structured linguistic system that must possess high narrative and logical cohesion. This explains my deep-seated preference for Spec-Driven Development, where software specifications are treated as semantic contracts governing automated execution.

Profile Attribute	Technical & Operational Specification
Legal Name	Soumyadeep Nath
Geographic Base	Kolkata, West Bengal, India
Contact Channels	soumyadeep9300@gmail.com github.com/sdn9300 sdn9300.github.io
Core Role Alignment	Aspiring Data Scientist, Full-Stack AI Builder, Forward Deployed Specialist
Academic Credentials	M.A. & B.A. (Honours) in English Literature, Presidency University, Kolkata
Technical Specialization	Executive PG Programme in Data Science & AI, IIT Roorkee (In Progress)
Target Roles	AI/ML Internships, Entry-Level Data Science & GenAI Roles
Primary Core Strengths	Interdisciplinary Translation, Spec-Driven Engineering, Robust Inference

Positioning & Credibility Principles:

- **Be honest about skill level and scope:** Declare credentials and capability transparently.
- **Prefer accurate framing over inflated titles:** Undersell rather than oversell; treat credibility as a long-term asset.
- **Dual-Advantage Framing:** Literature taught me to think clearly and communicate well. Data Science gives me analytical and technical leverage. Together, they let me build systems and explain them.
- **Academic & Tutoring Context:** Over a year of private tutoring experience (school-level English) involving data-driven student assessments, which anchors the focus on pedagogical structure.

2. Technical Profile & Tooling Taxonomy

My technical skill level is self-assessed as basic to early practical across Python and SQL, coupled with a deliberate strategy of utilizing AI-assisted construction to build production-grade systems as a primary learning vehicle. R serves as the primary computational engine for statistical analysis.

Deliberate AI-Assisted Engineering Strategy: There is no conflict between my basic solo coding level and the production-grade architectures shown in my portfolio. I deliberately choose to undertake advanced systems—featuring GLMM/Negative Binomial modeling, full CI/CD, Docker, and deterministic LLM guardrails—by working in tandem with AI assistance. This serves as an accelerated learning vehicle and portfolio strategy, shifting the learning curve from rote syntax acquisition to system orchestration, data flow engineering, and automated test-driven validation.

Tooling Domain	Core Technologies, Systems & Packages
Statistical Computing	R Language, Generalized Linear Mixed Models (GLMM), Negative Binomial Models, SciPy
General Programming	Python, TypeScript, Vanilla HTML/CSS/JavaScript
Web Architectures	Next.js 14 (App Router), Vite, Tailored Design Token Systems, Tailwind CSS
AI & LLM Integration	Groq API, Structured Prompts, Deterministic Hallucination Guardrails, Zod, LLaMA 3.3 70B
Data Scraping & Ingestion	StatsBomb API, FBref Parsers, BeautifulSoup4, Custom Ingestion Workers, Firecrawl API, Google Sheets API
Geospatial & Environmental	OpenStreetMap (OSM) Services, Open-Elevation API, Meteostat API
DevOps & Infrastructure	Docker Containerization, GitHub Actions CI/CD, pytest Suite Hardening, Redis, Prometheus, ReportLab, WakaTime
Deployment Targets	Vercel (Edge Runtime), GitHub Pages, Streamlit Community Cloud
Visual Customization	Shields.io (For-the-Badge styling), Capsule-Render API, Readme-Typing-SVG, Plotly, Streamlit

Active Learning Reference Resources:

Reference Resource	Material Type	Core Operational Application
Python for Data Analysis (Wes McKinney, 2nd Ed.)	Textbook	Structured pandas and NumPy operations
NumPy Cheat Sheet (Official)	Quick Reference	Vectorized mathematics, array manipulation
Pandas Cheat Sheet (Official / Irv Lustig)	Quick Reference	DataFrame manipulation, merging, aggregations
Python Beginner's Cheat Sheet	Quick Reference	Base syntax structure, control flows (Python Crash Course)
Google Colab & Jupyter Notebooks	Workspace	Interactive playground, Python experimentation & prototyping

3. Exhaustive Project Portfolio Analysis

The projects in my portfolio serve as live proof of technical execution, utilizing AI-assisted construction to build at production-grade standards.

Initiative Name	System Architecture	Primary Objective & Analytics Domain	Key Engineering Innovations
Altitude & Heat Adaptation	R, Python, Next.js 14, StatsBomb API, Meteostat, Open-Elevation, OSM	Sports Analytics, Applied Climatology & Inferential Statistics	GLMM Negative Binomial regression modeling, range-based projections with leave-one-tournament-out CV
AlignResume	Next.js 14, TypeScript, Groq API, Zod, Playwright/React-PDF	LLM Guardrails & Semantic Document Optimization	Deterministic guardrails preventing LLM generation of unverified credentials, Playwright proofs
Future Fit	Python, Prophet, mlxtend (Apriori), SciPy, Streamlit, Plotly, Groq API	Labor Market Skill Trend Forecasting & Association Mining	224K+ skill mentions from 50K+ postings; Prophet forecasting + Apriori co-occurrence rules + Skill Gap Advisor
Gleaner (Job Scout)	Python, BeautifulSoup4, Firecrawl API, Google Sheets API, YAML	Job-Listing Scraping & ETL Pipeline Integration	Adapter-pattern multi-source scraper (4+ boards) with dedup, keyword/location filters, YAML CLI
Overture	Python, Command-Line Interface, pytest, Docker	Automation Engineering & Test Hardening	Production-ready email worker verified with comprehensive unit testing (pytest) and Docker isolation

Project Details & Core Mechanics:

Altitude & Heat Adaptation:Models performance count outcomes using a GLMM with a Negative Binomial distribution in R. Math formulation:

$$\log(\mu_{ij}) = \beta_0 + \beta_1 \text{Altitude}_i + \beta_2 \text{ApparentTemperature}_{ij} + \beta_3 \text{TravelDistance}_{ij} + z_j b_j$$
where μ_{ij} represents the expected performance outcome of team i in match j , and b_j is a random intercept vector capturing tournament-specific variance. Projections are range-based conditional intervals validated using leave-one-tournament-out cross-validation. Employs a 37-scenario edge case registry and Go/No-Go gates.

AlignResume (The Resume Shapeshifter): Deployed at *align-resume-beta.vercel.app*. Features deterministic hallucination guardrails. Utilizes strict Zod schema validation rules to anchor the LLM's tailoring engine to user-provided source data. Final verified output is displayed side-by-side using Playwright and React-PDF.

Future Fit: Processes a dataset of 224,605 skill mentions from 50,000+ postings (2020-2026). Passes data through a 10-step cleaning and deduplication pipeline.

- 1. *Association rule mining:* maps 230 frequent skill itemsets via Apriori (lift ≥ 0.90 , confidence ≥ 0.38).
- 2. *Time-series forecasting:* Prophet projects skill share trajectories into 2027, with a fallback to linear extrapolation if the 80% confidence interval width exceeds a 1.5x variance threshold.

Gleaner (Job Scout): Multi-board scraper targeting Naukri, Indeed, RemoteOK, and Wellfound. Scraping resilience is maintained using stealth configuration headers, custom bot user-agents, and Firecrawl API's structured extraction schemas to bypass anti-bot mechanisms. Synchronizes live with Google Sheets API and local CSV channels.

4. Skill Blocks & Pedagogical Progress

To track my progression from basic scripting to advanced data engineering, my technical learning is structured into verified skill blocks.

Adaptive Tutoring Record: Hands-on programming challenges are performed using an Adaptive Coding Examination System. At ****Level 2****, completed *The Sales Report* challenge with a ****Perfect Score****, demonstrating mastery of Groupby, idxmax, and multi-conditional np.select() masking. At ****Level 3****, currently working on the *Student Grade Analyzer* challenge introducing float precision rounding (.round(n)), multi-conditional boolean masks, and frequency distributions (.value_counts()).

Pandas & NumPy Skill Block	Verified Status	Contextual Implementation Details
Groupby Aggregations	Mastered	Multi-variable grouping and aggregation functions (e.g. sum, mean, count)
Index Maxima Retrieval	Mastered	Utilizing idxmax() to isolate peak indexed values across grouped series
Label-Based Slicing	Mastered	Multi-axis index querying and row/column extraction via df.loc[]
Categorical Masking	Mastered	Multi-condition vector assignment with np.select()
Row Filtering & Querying	Introduced	Row-wise selection using compound logical boolean masks
Float Precision Manipulation	Introduced	Element-wise precision rounding using .round(n)
Categorical Distribution	Introduced	Frequency profiling via Series .value_counts()
Vectorized Lambda Transforms	Upcoming	Vector mapping with .apply() and anonymous lambda functions
Multi-Index Reshaping	Upcoming	Multi-dimensional reshaping via .pivot_table(), melt(), and stack()
Relational Merging	Upcoming	Executing explicit tabular joins (inner, outer, left, right), concatenations
Datetime Indexing	Upcoming	Manipulating datetime objects for time-series parsing (Future Fit)

NumPy Comprehensive Lab Details:

Status: ****Completed with a Perfect Score (15/15)****.

NumPy Topic Block	Subtopics & Core Operational Functions Tested
Array creation	np.array(), np.zeros(), np.ones(), np.arange(), np.linspace(), np.random.rand(), np.random.randint(), np.random.randn()
Array properties	.shape, .dtype, .ndim, .size
Indexing & Slicing	1D indexing, 2D [row, col] indexing, negative indexing, arr[start:stop:step], submatrix slicing
Vectorized math & Stat	+, -, *, /, **, np.sqrt(), np.exp(), np.log(), np.sum(), np.mean(), np.std(), np.min(), np.max() with axis parameter
Broadcasting & Reshaping	Scalar-array and compatible shape operations, .reshape(), .flatten(), .ravel()
Boolean indexing & ufuncs	arr[arr > value], compound boolean conditions, np.sqrt, np.abs, np.ceil, np.floor

Verified Code Patterns Written in actual sessions:

- Groupby + aggregation: df.groupby('region')['revenue'].sum()
- Finding index of max: best_idx = df.groupby('region')['revenue'].idxmax()
- Lookup with loc: df.loc[best_idx, 'product']
- Multi-condition columns: df['grade'] = np.select(conditions, choices, default='F')

5. Workflow Preferences & Spec-Driven Development

I operate strictly under Spec-Driven Development (SDD) principles. No code is written before a specification exists.

The operational mechanics of SDD are executed through the open-source GitHub Spec Kit framework, following a structured 6-phase lifecycle: 1. CONSTITUTION (speckit.constitution), 2. SPECIFY (speckit.specify), 3. PLAN (speckit.plan), 4. TASKS (speckit.tasks), 5. IMPLEMENT (speckit.implement), 6. VALIDATE & REPLAN (pytest / Model Evaluation).

Specification Rigor Level	Primary Core Objective	Core Artifact Location	Alignment with Soumyadeep's Stack
Spec-First	Establishes early alignment; spec not maintained post-code generation	Initial Prompt / Disposable Doc	Used for rapid, low-complexity prototyping tasks
Spec-Anchored	Specification evolves alongside codebase as a living reference	Repository /docs or /specs	Primary standard for flagship systems and statistical pipelines
Spec-as-Source	Direct code generation driven from the spec; humans do not edit code	Config/DSL File & Compiler	Restricted to isolated, low-criticality automation tasks

Operating Guidelines:

- **Documentation First:** Explain why choices are made, not just what the code does. ADRs are preferred over terse output.
- **Root-Cause Debugging:** Never apply quick patches or surface-level band-aids. Identify and fix the actual source cause.
- **Collaborator Behavioral Expectation:** AI assistants must act as senior advisors, architects, and researchers. Verify design decisions against the spec before proposing code changes.

6. Pedagogical & Learning Frameworks

My learning style is highly structured, adaptive, and pedagogical. I respond best to 'intuition-first' explanations.

General Coding Support System Modes:

Operational Mode	Mode Label	Functional Objective
MODE 1	EXPLAIN	Provide step-by-step concept explanations with architectural trade-offs
MODE 2	PROBLEM GENERATOR	Generate industry-aligned coding problems with defined constraints
MODE 3	SOLUTION WALKTHROUGH	Trace logic from brute force to optimized code, analyzing complexity
MODE 4	DEBUGGER	Execute root-cause analysis, tracing bugs to fix source logic
MODE 5	PRACTICE SYSTEM	Deliver structured, progressive problem sets to build target skills
MODE 6	DOUBT SOLVER	Resolve specific syntax queries, analyzing engineering trade-offs

Quantitative Skill Acquisition Status:

Difficulty Level	Complexity Label	Operational Status
Level 1	Easy	Completed successfully
Level 2	Easy-Medium	Completed with a perfect score (verified)
Level 3	Medium	In Progress (Grade Analyzer module)
Level 4	Medium-Hard	Upcoming pedagogical milestones
Level 5	Hard	Target performance milestone

7. Adaptive Coding & Concept Contracts

Adaptive Coding Examiner (CAT System): My programming practice sessions scale difficulty dynamically. A fully correct implementation increases difficulty by 1 level; a logical failure drops it by 1 level; a partially correct solution holds the current level. AI collaborators must follow a strict flow: Assessment -> Delivery (no solution shown) -> User Attempt -> Evaluation (O(N) time/space analysis) -> Feedback -> Adjustment -> Iteration.

Concept Explanation Contract: Concept breakdowns must contain six parts in sequence:

1. **Concept:** Define the core idea or algorithm plainly.
2. **Breakdown:** Provide a step-by-step, chronological explanation of mechanics.
3. **Analogy:** Use a real-world physical analogy to ground abstract logic.
4. **Visual Enhancement:** Construct structured ASCII diagrams or charts.
5. **Why It Matters:** Explain real-world application, system impact, and architectural trade-offs.
6. **Mini Check:** Provide a single-question self-test with answers immediately following.

Required Closing Sequence: Every tutoring session must conclude with:

Final Summary -> Key Takeaways (4-6 bullets) -> Knowledge Check Quiz (6-8 questions without answers shown) -> Answer Key (one-sentence explanations for self-grading) -> Feedback Loop invitation.

8. Cognitive, Behavioral & Communication Profiles

Trait	Behavioral Evidence	Operational Strategic Alignment
Precision-Oriented	Writes explicit compound logic; rejects hand-waving	Reduces code logic errors; builds robust pipelines
Disciplined Planner	Prepares 37-scenario edge case tests prior to coding	Prevents code drift and system architectural regressions
Documentation-First	Maintains thorough /specs directories and ADR records	Ensures long-term maintainability for collaborative dev
Intellectually Honest	Declares credentials accurately; avoids inflated claims	Builds long-term professional credibility as an asset
Literature-Trained	Writes dense narrative prose; high semantic clarity	Bridges the gap between complex stats and business logic
Aesthetically Intentional	Applies design tokens and professional UI styling	Creates production-grade presentation layers (Next.js)

Structured Communication Guidelines:

Communication Parameter	Authorized Formats & Styles	Restricted Words & Patterns	Engineering Rationale
Tonal Paradigm	Academic, authoritative, pedagogical, mathematically precise, logically verified	Casual, conversational, overly simplified, hand-waving, speculative	Ensures technical precision and conceptual depth
Information Layout	Detailed markdown tables, continuous prose, schema contracts, LaTeX equations	Bulleted qualitative paragraphs, raw unformatted code blocks, sparse summaries	Promotes readability and clear data structure mappings
Engineering Rationale	Documenting the 'why' of choices, explicit trade-offs, and Architecture Decision Records (ADRs)	Pure syntax dumps, generic auto-generated comments, undocumented features	Establishes system maintainability and historical context
Terminology Standards	Strict statistical, mathematical, and computer science nomenclature	Hyperbolic marketing terms, vague software adjectives, unverified assertions	Maintains academic and technical rigor

9. Career Direction, Living Notes & Update Log

Resume Version Control & Sourcing Workflows:

- **Programmatic PDF Generation:** Resume is generated dynamically via Python + ReportLab. The current iteration is **v12** (12+ rounds of revision).
- **Rigorous Evidence Audit:** Removed unevidenced terms (Shadcn UI, OpenAI API, Power BI, Feature Engineering, Google Colab) to preserve credential honesty. Added Playwright since it is explicitly supported by AlignResume.
- **Structural Fixes:** IIT Roorkee moved to Education & Training; private tutoring gap filler deliberately removed since three live projects are sufficient; relabeled portfolio links.
- **Job Search Sourcing:** Primary query executions via parameterized channels (Indeed API queries) targeting Kolkata and remote roles (country_code: IN) in Data Science, GenAI, Data Analyst, NLP, using qualifiers like 'junior' or 'fresher'. Target sectors: EdTech, IT Services, Media/Content-Tech.
- **Surfaced Sourcing Targets (June 2026):** Gen AI Engineer @ Encoders, Senior Data Analyst/AI Developer @ Web Spiders, AI/ML Gen AI (RAG) Engineer @ Webskitters, AI/ML Engineer @ Accenture.

Profile & Sourcing Roadmap:

- **Phase 1 (Immediate Alignment):** Correct metadata across public repos (standardize titles, write READMEs, link deployment URLs, remove stray .agents/ artifacts).
- **Phase 2 (Data Science Core):** Refine statistical pipelines and R scripting reproducibility files.
- **Phase 3 (Agentic Systems):** Build and catalog robust LLM orchestration nodes and pytest suites.
- **Phase 4 (FDE Integration):** Aggregate multi-system orchestrations and deployment architectures.

Appendix A: Onboarding Identity Card (Fast Context):

Soumyadeep Nath (Kolkata, India) | Executive PG in DS & AI (IIT Roorkee, in progress) + M.A. in English Lit (Presidency University) | Track: Aspiring Data Scientist -> AI Engineer -> Forward Deployed Engineer | Stack: Python (Pandas, NumPy), R (GLMM, Negative Binomial), TypeScript (Next.js 14) | Behavior: Spec-first, root-cause debugger, hint-first learner | Target Sectors: EdTech, IT Services, Media/Content-Tech.

Update Log & Living Document Notes:

- Added explicit teaching, coding preference, and quiz formatting contracts.
- Incorporated project details, portfolio principles, and tech stacks from recent work.
- Codified career path transition: Literature -> Data Science -> AI Engineer -> FDE.
- This document serves as a living, consolidated profile to be shared with new AI systems and collaborators to prevent context drift.